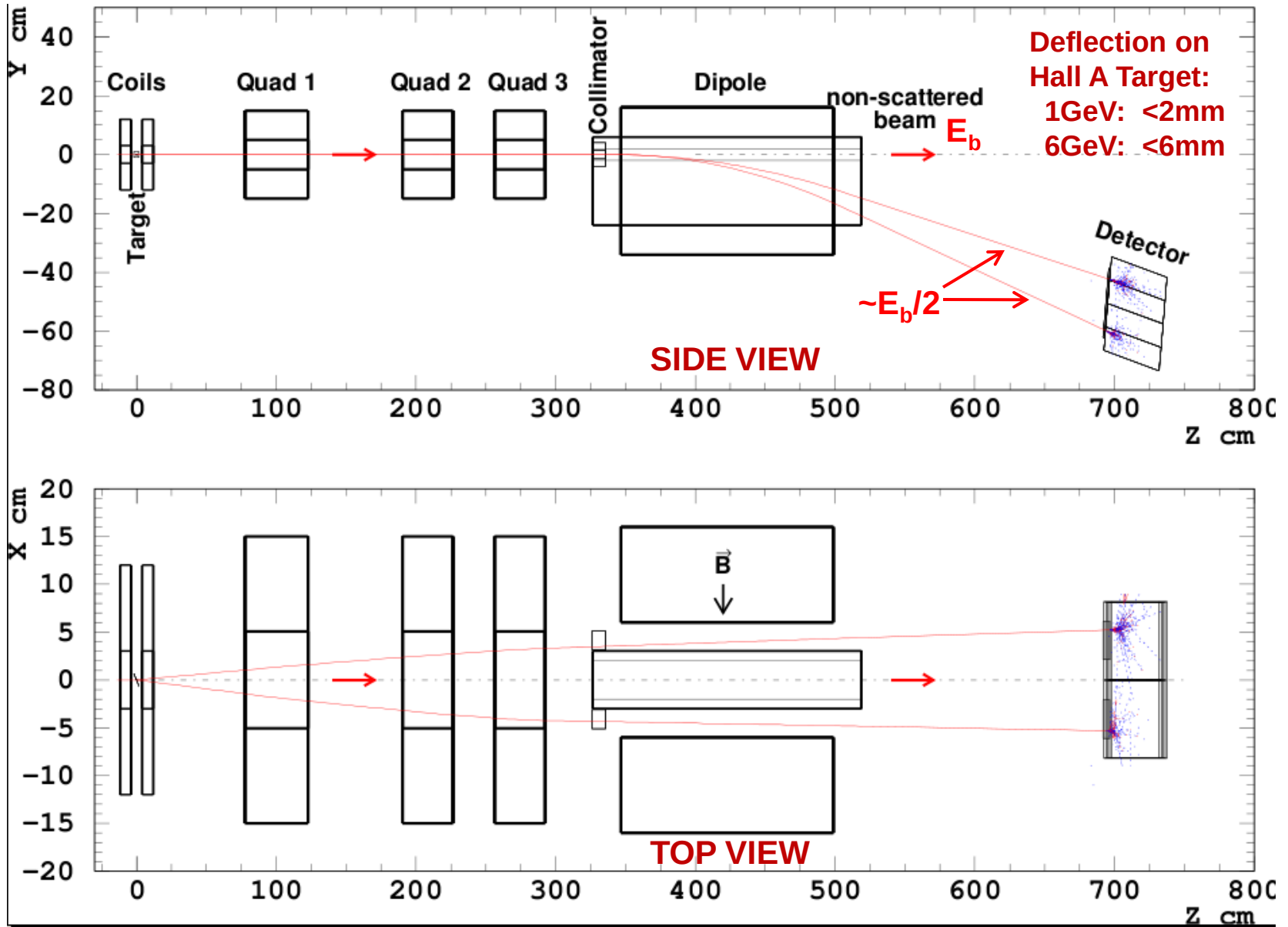


Hall A Møller Polarimeter after Upgrade

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Møller Polarimeter Layout



Møller Polarimetry

$$\vec{e}_b^- + \vec{e}_t^- \rightarrow e_{Moller}^- + e_{Moller}^-$$

$$A_{meas} = \frac{N_+ - N_-}{N_+ + N_-} = AnPow \bullet P_t \bullet P_b$$

- A_{meas} – measured asymmetry;
 N_+/N_- – Møller detector counting rates for helicity +/-;
 $AnPow$ – analyzing power (from GEANT), **depends of beam position**;
 P_t – Møller target polarization (from literature);
 P_b – beam polarization

$$P_b = \frac{A_{meas}}{AnPow \bullet P_t}$$

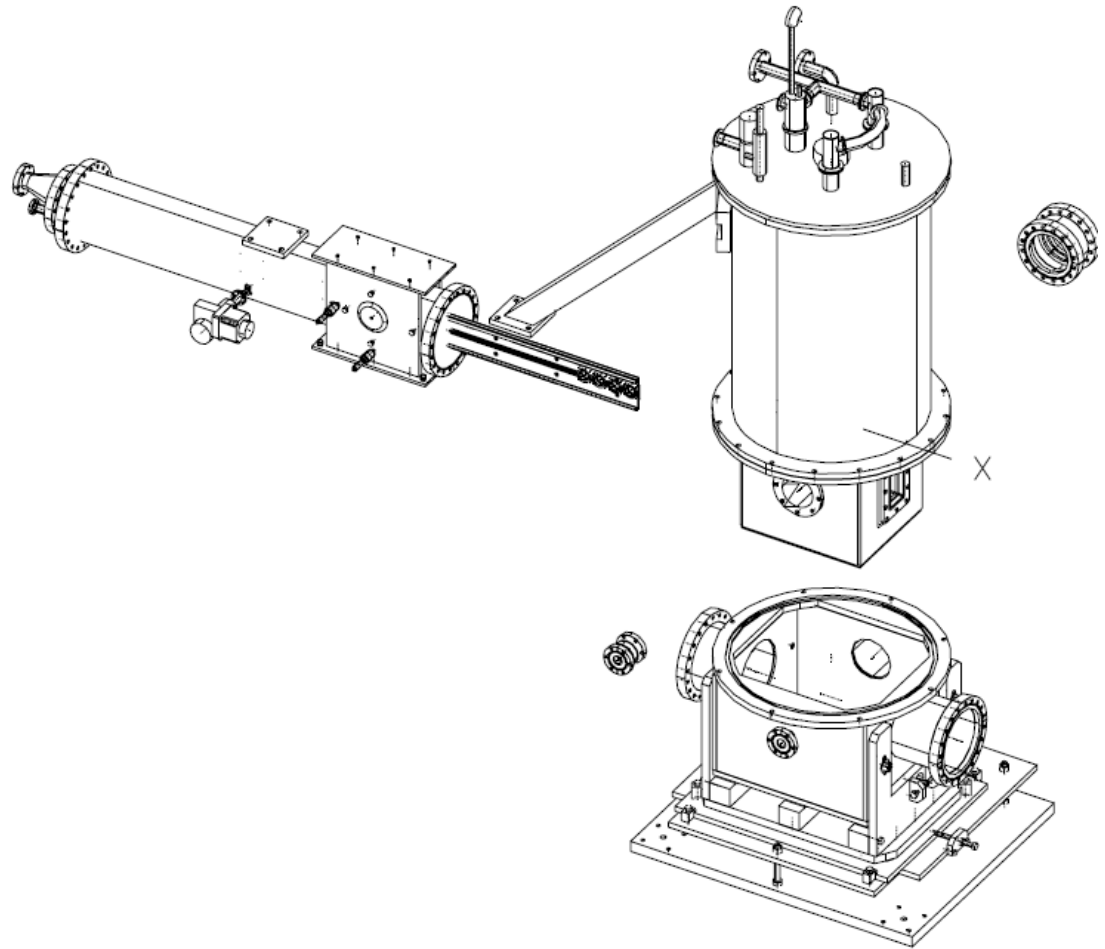
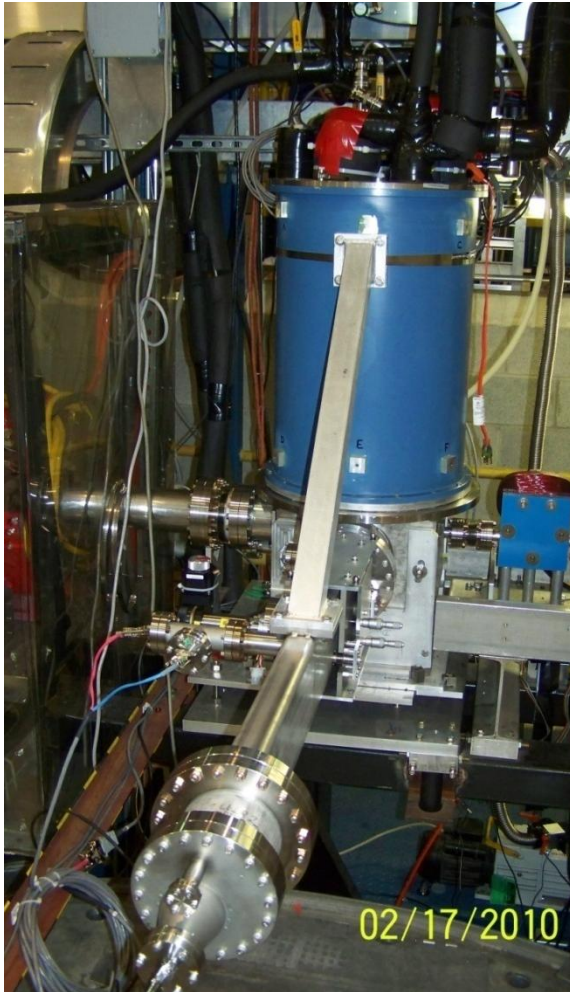
Why We Need Upgrade?

Møller polarimeter now:

- the beam current $0.3 \div 1 \mu\text{A}$;*
- closed slit (slit position from 14 to 18 r.u.);*
- no energy/position fast feed-back*

- 1. Polarized electron target "brute force" 3T
(100% target magnetization);*
- 2. Introduce a beam duty cycle:
reduce target heating, increase beam current:*
 - no laser phase dependence (fully open slit);*
 - energy fast feed-back;*
 - beam position fast feed-back;*
- 3. New fast DAQ based on FADC (higher rate)*
- 4. Segmented aperture detector (higher rate)*

New Møller Target Design

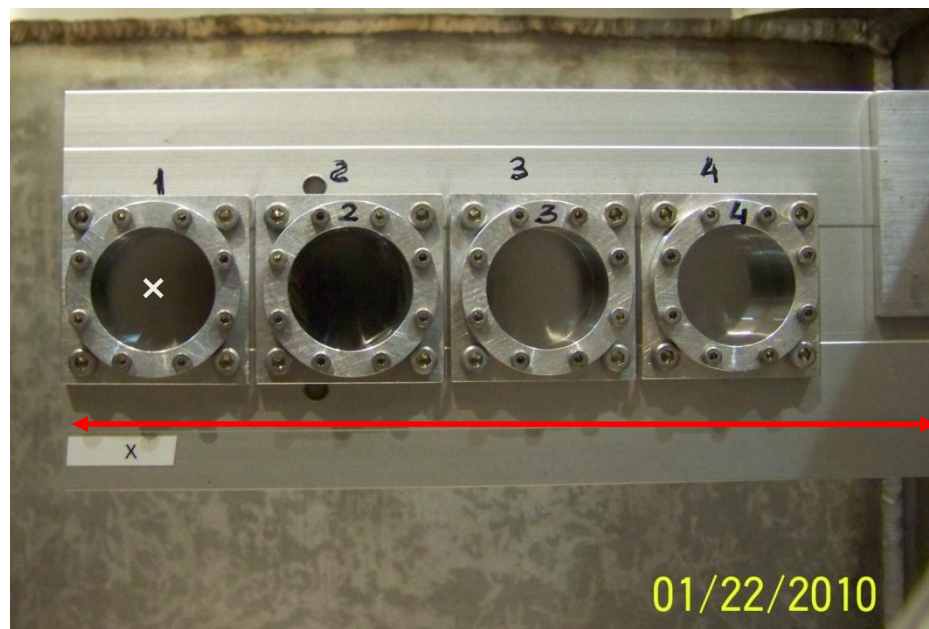
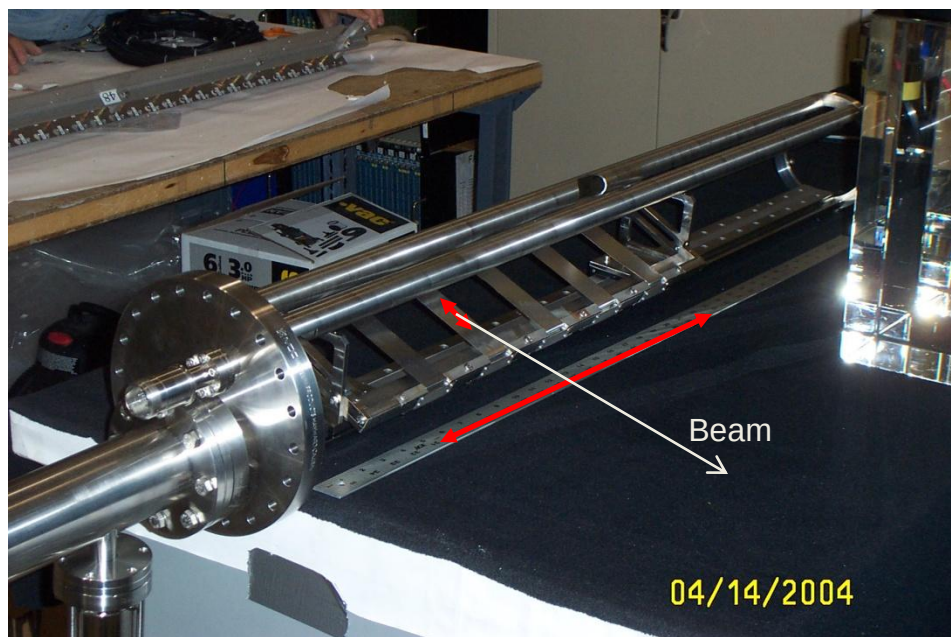


*Superconducting Helmholtz Coils ~3 T magnetic field
For PREX: $\sim 7^\circ$ of the beam XY rotation
~25 min to rump up/down*

Møller Target Holder

Old Target Holder

New Target Holder



5 target foils:

SM-7 μ m, Fe-7 μ m, Fe-14 μ m,

SM-29 μ m, SM-13 μ m

SM- Supermendur (49%Fe 49% Co2%V)

Tilted 20° to the beam direction

Moved along the beam and

in transverse to the beam direction

Can be moved with the beam ON

4 target foils:

Fe-1 μ m, Fe-1 μ m, Fe-4 μ m, Fe-12 μ m

Tilted 90° to the beam direction

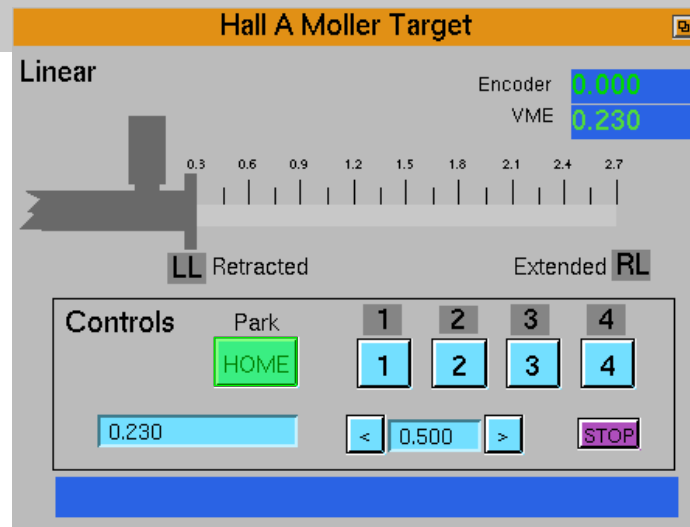
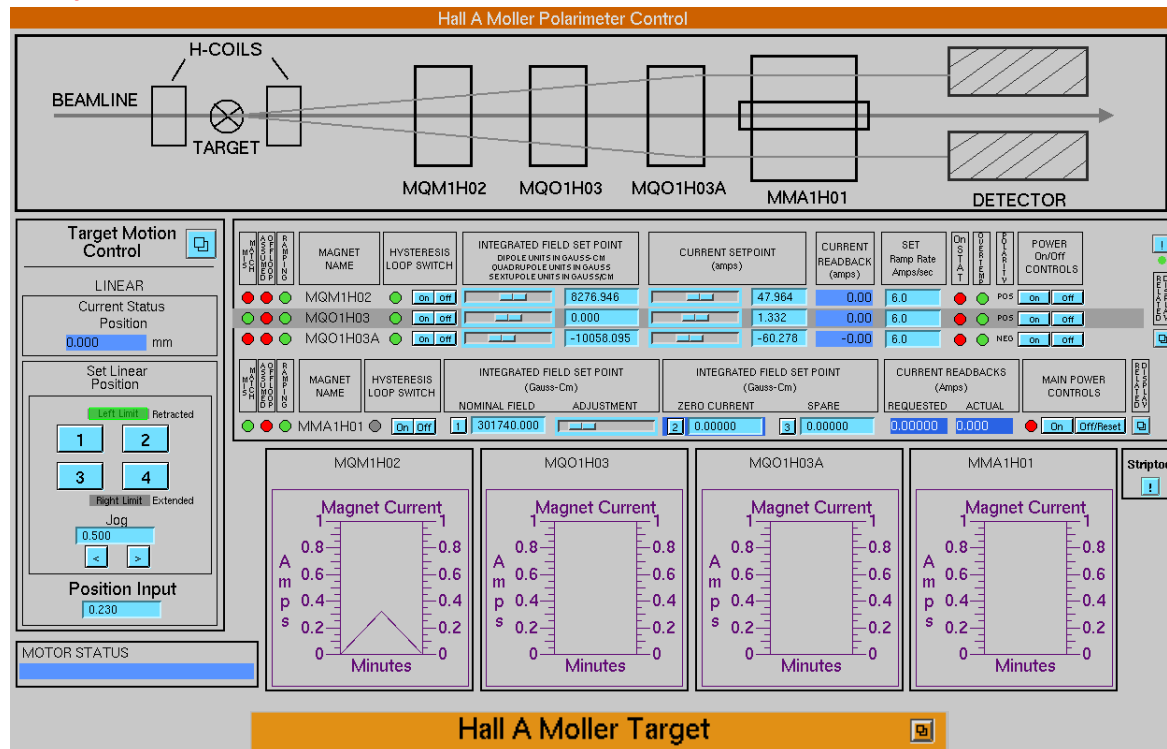
Moved in transverse to the beam

direction

The beam **MUST BE OFF** for any target movement

Only MCC can move the target

Updated MCC Møller screens



Møller Requirements for the Beam

- *Fast Raster $1.5 \times 0.5 \text{ mm}^2$ (PREX with Møller sett.)
 $1 \times 0.5 \text{ mm}^2$ (for Møller measurements)*
- *Beam position:
centered $XY \pm 0.2 \text{ mm}$ (BPM1HO1);
parallel (the best tuning);*
- *Measurements with **CW** beam up to $5 \mu\text{A}$;*
- ***"EXPERT" mode** (invaizive?)
beam is ON part time of helicity window;
beam current up to $15 \mu\text{A}$;
fully open slit (use attenuator for current);*
- ***Beat Frequency mode** (for special test):
beam current up to $50 \mu\text{A}$;
micro-suppression $n=10$ (50MHz)
Møller procedure will be updated!!!*

"EXPERT" mode

Expert Beam Mode Control

Gun Beam Mode

Source Status: **Pol Src Ready**

Select Accel Mode: **BEAM SYNC ONLY**

Viewer Safe

Select A Laser Mode: **BEAM SYNC ONLY**

Duty Factor: 0.00 %

Viewer Safe

Select B Laser Mode: **BEAM SYNC ONLY**

Duty Factor: 0.00 %

Viewer Safe

Select C Laser Mode: **BEAM SYNC ONLY**

Duty Factor: 0.00 %

Viewer Safe

Beam Sync Status: **ON LINE SYNC** 60 HZ 60 Hz

Beam Sync ON
SCAM BEAM SYNC

Viewer Limited Settings

VL Pulse Width Set: 10.000 us

VL Pulse Width Read: 10.000 us

Modulation Width Read: 99.500 us

CW Checks

CW Crew Chief Permissive

CW ENABLE

CW INHIBIT

CW ENABLE

Hall A: **A ENABLE**, **A INHIBIT**

Hall B: **B ENABLE**, **B INHIBIT**

Hall C: **C ENABLE**, **C INHIBIT**

CW ENABLE

CW MSlt/C.Plug Permissive

Override ON, **Override OFF**

CW ENABLE

CW Hall/Slit Permissive

Override ON, **Override OFF**

CW ENABLE

When Expert is selected on Beam Mode beam sync will be set by the Mizar instead of the SCAM. In this mode the Lasers will only allow operation in Expert or Beam Sync modes.

MIZAR EXPERT GUN CONTROL

Gun Beam Mode

Select Beam Mode: **BEAM SYNC ONLY**

Beam Mode Status: **Beam Sync Mode**

Beam Parameters: **No FSD**

Duty Factor: 0.00 %

Source Status: **Pol Src Ready**

Switch Sources

Beam Envelope Control

Viewer Limited Settings

VL Pulse Width: 0.20 us

VL Pulse Period: 16666.67 us

VL Pulse Frequency: 60.00 Hz

Status: **Setting Accepted**

Viewer Safe

Pulse Mode Settings

Pulse Width: 10.00 us

Pulse Period: 16666.67 us

Pulse Frequency: 60.00 Hz

Status: **Setting Accepted**

CW Mode Settings

Pulse Width: 16666.67 us

Pulse Period: 16666.67 us

Pulse Frequency: 60.00 Hz

DC ON Control: **OFF**, **ON**

Status: **May Lose Sync or Beam Pulses**

Clock and Delay Control

Select Reference Clock: **SCAM SYNC**

Select Line Sync Frequency: **60 HZ**

Set Delays

Clock Ref to Beam Sync Delay: 1.00 us

Beam Sync To Beam Delay: 0.00 us

Beam Sync To Mod Delay: 1388.89 us

Modulation Mode Settings

Number of Mod Pulses: 4

Modulation Width: 2777.78 us

Modulation Low: 0.10 us

Modulation Period: 16666.67 us

Modulation Frequency: 60.00 Hz

Status: **Setting Accepted**

Beam Sync Settings

Beam Sync Width: 16666.67 us

Beam Sync Period: 16666.67 us

Beam Sync Frequency: 60.00 Hz

Status: **May Lose Beam Sync Pulses**

"EXPERT" mode - kind of a pulse beam

New Møller FSD

Fast Shutdown System - Moeller Target Masking

The Question: Should we mask the Hall A Moeller Target FSD?

Target Mask Request Request Readback

Target UNMask Request

-1.00 Slit Value iocin1:heartbeat 182336

0.00 BLA Current iocbcm:heartbeat 706975

ISD1H001L.H (Moeller Mask Request)

ISD1H001NA.B7 (Moeller Mask Hardware Readback)

The Judges

				
Yea	Yea	Nay	Yea	Yea
iocin1 Heartbeat	iocbcm Heartbeat	Slit A Status	Hall A BCM Status	All Sequence Channels Connected
C	D	A	B	E



Masking OFF!
The Decision

Application is fsdwt (wft = waterfall target) -- also see the fsd application.

The Moeller Target can be moved without an FSD trip, only if the Moeller FSD is masked.

The Moeller FSD can be masked only from this screen via the above [!] Target Mask Request button.

Masking is allowed only when all 5 Judges are green (Yea).

The Slit A Judge is green only for a slit position between 8 and 30.

The Hall A BCM Judge is green only for a beam current of less than 5 uA.

OLD
SCREEN
will be
updated

CW mode - FSD the same

USER mode - slit fully open, beam current limit 60μA

Hall A will never request masking of Møller target movement

Time Budget for Møller Measurement

Møller settings (+magnet ramp up): ~1.5hour

Raster $1 \times 0.5 \text{mm}^2$

CW: ~2hours

$1 \mu\text{m} / 3 \mu\text{A} \sim 1 \text{hour}$ (0.3%stat error)

$4 \mu\text{m} / 3 \mu\text{A} \sim 20 \text{min}$ (0.3%stat error)

$12 \mu\text{m} / 1 \mu\text{A} \sim 20 \text{min}$ (0.3%stat error)

Target thickness and condition systematic

"EXPERT" mode: ~1hour

10ms of 30ms helicity pulse (30Hz) or

1.3ms of 4ms helicity pulse (240Hz)

$1 \mu\text{m} / 10 \mu\text{A}$ beam: $\sim 45 \text{min}$ for 0.3% stat.er.

Restore PREX beam settings: ~0.5hour

Total: ~5hours